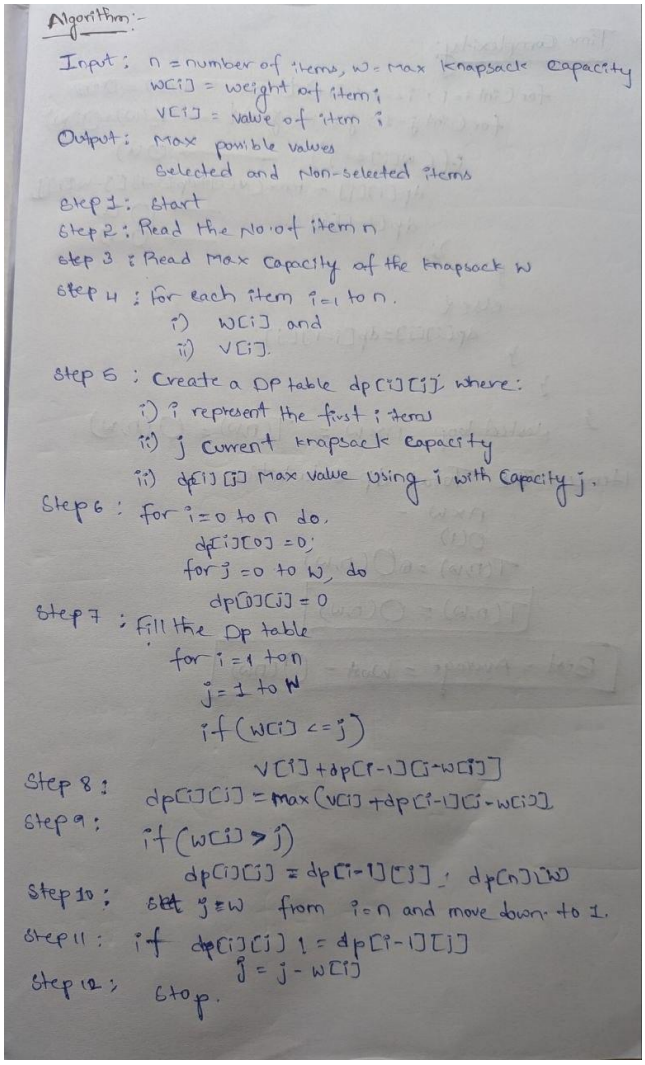
Practical: 5	Implementation of a knapsack problem using dynamic programming.
25/08/2026	

Aim: To Implement the knapsack problem using dynamic programming.

Algorithm:



Program:

```
#include<iostream>
using namespace std;

class Knapsack {

private:
   int w[10];       // weights
```

```
    int v[10];      // values
    int n;          // number of items
    int W;          // capacity
    int dp[10][100]; // dp table

public:

    // ?? Get Input ??
    void getInput() {
        cout << "Enter number of items   : ";
        cin  >> n;
        cout << "Enter knapsack capacity : ";
        cin  >> W;
        for(int i = 1; i <= n; i++) {
            cout << "Item " << i << " weight and value : ";
            cin  >> w[i] >> v[i];
        }
    }

    // ?? Build dp table ??
    void solve() {

        // base case - row 0 and col 0
        for(int i = 0; i <= n; i++) dp[i][0] = 0;
        for(int j = 0; j <= W; j++) dp[0][j] = 0;

        // fill table
        for(int i = 1; i <= n; i++) {
            for(int j = 1; j <= W; j++) {

                if(w[i] <= j) {
                    // take or not take
                    dp[i][j] = max(
                        v[i] + dp[i-1][j-w[i]],
                        dp[i-1][j]
                    );
                } else {
                    // cannot take item i
                    dp[i][j] = dp[i-1][j];
                }
            }
        }
    }

    // ?? Print dp table ??
    void printTable() {
        cout << "\nDP Table:" << endl;
```

```
        cout << "     ";
        for(int j = 0; j <= W; j++)
            cout << j << "  ";
        cout << endl;

        for(int i = 0; i <= n; i++) {
            cout << "i=" << i << "  ";
            for(int j = 0; j <= W; j++)
                cout << dp[i][j] << "  ";
            cout << endl;
        }
    }

    // ?? Find selected items ??
    void findItems() {
        cout << "\nSelected Items:" << endl;
        int j = W;
        for(int i = n; i >= 1; i--) {
            if(dp[i][j] != dp[i-1][j]) {
                cout << "Item " << i
                     << " (w=" << w[i]
                     << ", v=" << v[i]
                     << ") TAKEN " << endl;
                j = j - w[i];
            } else {
                cout << "Item " << i
                     << " (w=" << w[i]
                     << ", v=" << v[i]
                     << ") NOT taken " << endl;
            }
        }
    }

    // ?? Show result ??
    void showResult() {
        cout << "\nMaximum Value = "
             << dp[n][W] << endl;
    }
};

// ?? MAIN ??
int main() {

    Knapsack k;

    k.getInput();
    k.solve();
```

```
    k.printTable();
    k.showResult();
    k.findItems();

    return 0;
}
```

Output:

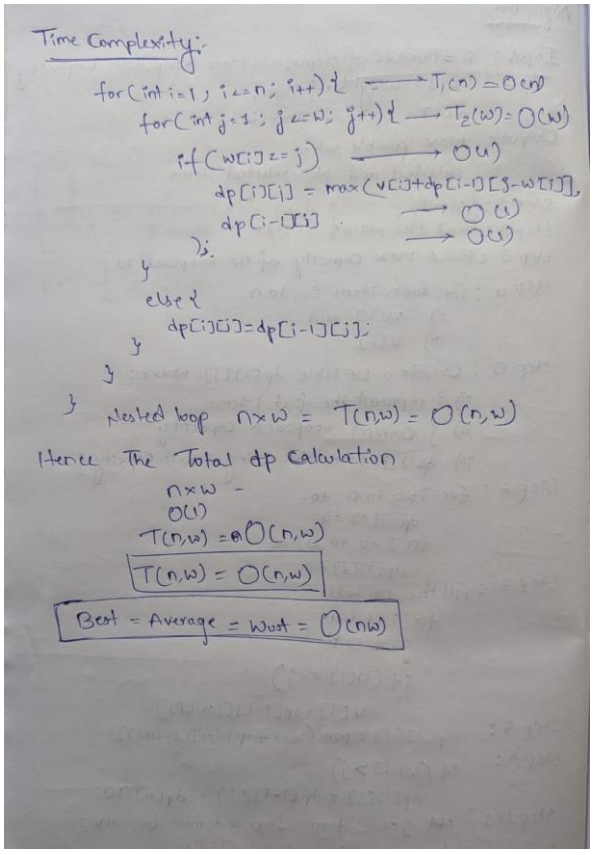
```
C:\Users\Chandu\OneDrive\D
Enter number of items    : 2
Enter knapsack capacity : 4
Item 1 weight and value : 2 4
Item 2 weight and value : 6 4

DP Table:
      0  1  2  3  4
i=0   0  0  0  0  0
i=1   0  0  4  4  4
i=2   0  0  4  4  4

Maximum Value = 4

Selected Items:
Item 2 (w=6, v=4) NOT taken
Item 1 (w=2, v=4) TAKEN
```

Time Complexity:



Conclusion:

The Implementation of the knapsack problem using dynamic programming successfully executed**.**

And the time complexity analysis is same for all the three cases Best case = Average case = Worst Case = O(n,w)

Where n is Number of items and w is the Maximum capacity of the knapsack.